DSA0503 – Query Processing for Data Science

Experiments 11–20

EXPERIMENT 11

Question: Create a dataframe of ten rows, four columns with random values. Write a Pandas program to convert some values to NaN values and highlight the NaN values.

Program / Code

```
import pandas as pd
import numpy as np

df = pd.DataFrame(
np.random.randn(10, 4),
columns=["A", "B", "C", "D"]
)

# Convert some values to NaN
df.iloc[1, 2] = np.nan
df.iloc[4, 0] = np.nan
df.iloc[7, 3] = np.nan

def highlight_nan(value):
if pd.isna(value):
return "background-color: yellow"
return ""

df.style.map(highlight_nan)
```

Output

	A	B	C	D
0	-2.248921	-0.587322	0.412002	1.775217
1	-0.791237	0.539388	nan	-0.241929
2	-0.086298	-0.230184	-1.290793	-0.555702
3	0.337074	1.076709	-0.620083	-1.106450
4	nan	-0.181732	1.875593	-1.511679
5	0.023022	-0.323631	-0.355974	-1.295517
6	0.140395	-0.489184	2.981822	-0.731082
7	1.209580	1.018186	-0.607671	nan
8	0.319030	-1.027495	-0.700703	-1.093716
9	-1.435073	-0.442806	0.204899	0.733891

EXPERIMENT 12

Question: Create a dataframe of ten rows, four columns with random values. Write a Pandas program to set dataframe background color black and font color yellow.

Program / Code

```
import pandas as pd
import numpy as np

df = pd.DataFrame(
np.random.randn(10, 4),
columns=["A", "B", "C", "D"]
)

df.style.set_properties(
**{
"background-color": "black",
"color": "yellow"
}
)
```

Output

	A	B	C	D
0	-1.069824	0.892747	-0.625006	-0.064588
1	-0.544045	0.807263	0.087328	-1.053289
2	0.885337	-1.160900	-1.418681	0.675143
3	-0.323427	1.011184	0.880460	-1.409149
4	-0.507148	-0.338938	0.638883	-0.708021
5	0.150695	0.674830	0.032939	0.198685
6	1.141754	-1.655475	0.850190	0.347660
7	-0.521630	-1.106904	-1.829678	-0.486784
8	-0.581290	0.118940	-1.059282	-0.905586
9	0.270050	-0.691741	2.100893	1.151917

EXPERIMENT 13

Question: Write a Pandas program to detect missing values of a given DataFrame. Display True or False.

Program / Code

```
import pandas as pd
import numpy as np

df = pd.DataFrame({
"A": [10, 20, np.nan, 40],
"B": [50, np.nan, 70, 80],
"C": [90, 100, 110, np.nan]
})

print(df.isnull())
```

Output

```
       A      B      C
0  False  False  False
1  False   True  False
2   True  False  False
3  False  False   True
```

EXPERIMENT 14

Question: Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

Program / Code

```
import pandas as pd
import numpy as np

df = pd.DataFrame({
"ord_no": [70001, np.nan, 70002, 70004, np.nan, 70005,
np.nan, 70010, 70003, 70012, np.nan, 70013],
"purch_amt": [150.50, 270.65, 65.26, 110.50, 948.50, 2400,
5760, 1983.43, 2480.40, 250.45, 75.29, 3045.60],
"ord_date": [
"2012-10-05", "2012-09-10", np.nan, "2012-08-17",
"2012-09-10", "2012-07-27", "2012-09-10",
"2012-10-10", "2012-10-10", "2012-06-27",
"2012-08-17", "2012-04-25"
],
"customer_id": [3002, 3001, 3001, 3003, 3002, 3001,
3001, 3004, 3003, 3002, 3001, 3001],
"salesman_id": [5002, 5003, 5001, np.nan, np.nan, 5002,
5001, np.nan, 5003, 5002, 5003, np.nan]
})

print("Original DataFrame:")
print(df)

df = df.fillna(0)

print("\nAfter replacing missing values:")
print(df)
```

Output

```
Original DataFrame:
     ord_no  purch_amt    ord_date  customer_id  salesman_id
0   70001.0     150.50  2012-10-05         3002       5002.0
1       NaN     270.65  2012-09-10         3001       5003.0
2   70002.0      65.26         NaN         3001       5001.0
3   70004.0     110.50  2012-08-17         3003          NaN
4       NaN     948.50  2012-09-10         3002          NaN
5   70005.0    2400.00  2012-07-27         3001       5002.0
6       NaN    5760.00  2012-09-10         3001       5001.0
7   70010.0    1983.43  2012-10-10         3004          NaN
8   70003.0    2480.40  2012-10-10         3003       5003.0
9   70012.0     250.45  2012-06-27         3002       5002.0
10      NaN      75.29  2012-08-17         3001       5003.0
11  70013.0    3045.60  2012-04-25         3001          NaN

After replacing missing values:
     ord_no  purch_amt    ord_date  customer_id  salesman_id
0   70001.0     150.50  2012-10-05         3002       5002.0
1       0.0     270.65  2012-09-10         3001       5003.0
2   70002.0      65.26           0         3001       5001.0
3   70004.0     110.50  2012-08-17         3003          0.0
4       0.0     948.50  2012-09-10         3002          0.0
5   70005.0    2400.00  2012-07-27         3001       5002.0
6       0.0    5760.00  2012-09-10         3001       5001.0
7   70010.0    1983.43  2012-10-10         3004          0.0
8   70003.0    2480.40  2012-10-10         3003       5003.0
9   70012.0     250.45  2012-06-27         3002       5002.0
10      0.0      75.29  2012-08-17         3001       5003.0
11  70013.0    3045.60  2012-04-25         3001          0.0
```

EXPERIMENT 15

Question: Write a Pandas program to keep the rows with at least 2 NaN values in a given DataFrame.

Program / Code

```
import pandas as pd
import numpy as np

df = pd.DataFrame({
"A": [1, np.nan, 3, np.nan, np.nan],
"B": [np.nan, 2, 3, np.nan, 5],
"C": [np.nan, np.nan, 3, 4, np.nan],
"D": [4, 5, np.nan, np.nan, np.nan]
})

print("Original DataFrame:")
print(df)

result = df[df.isna().sum(axis=1) >= 2]

print("\nRows having at least 2 NaN values:")
print(result)
```

Output

```
Original DataFrame:
     A    B    C    D
0  1.0  NaN  NaN  4.0
1  NaN  2.0  NaN  5.0
2  3.0  3.0  3.0  NaN
3  NaN  NaN  4.0  NaN
4  NaN  5.0  NaN  NaN

Rows having at least 2 NaN values:
     A    B    C    D
0  1.0  NaN  NaN  4.0
1  NaN  2.0  NaN  5.0
3  NaN  NaN  4.0  NaN
4  NaN  5.0  NaN  NaN
```

EXPERIMENT 16

Question: Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object.

Program / Code

```
import pandas as pd

data = {
"school": ["S1", "S2", "S3", "S4", "S5", "S6"],
"school_code": ["s001", "s002", "s003", "s001", "s002", "s004"],
"class": ["V", "V", "VI", "VI", "V", "VI"],
"name": [
"Alberto Franco", "Gino Mcneill", "Ryan Parkes",
"Eesha Hinton", "Gino Mcneill", "David Parkes"
],
"age": [12, 13, 13, 13, 14, 12]
}

df = pd.DataFrame(data)

grouped = df.groupby("school_code")

print("Type of GroupBy object:")
print(type(grouped))

print("\nGroups:")
for school, group in grouped:
print("\nSchool Code:", school)
print(group)
```

Output

```
Type of GroupBy object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>

Groups:

School Code: s001
  school school_code class            name  age
0     S1        s001     V  Alberto Franco   12
3     S4        s001    VI    Eesha Hinton   13

School Code: s002
  school school_code class          name  age
1     S2        s002     V  Gino Mcneill   13
4     S5        s002     V  Gino Mcneill   14

School Code: s003
  school school_code class         name  age
2     S3        s003    VI  Ryan Parkes   13

School Code: s004
  school school_code class          name  age
5     S6        s004    VI  David Parkes   12
```

EXPERIMENT 17

Question: Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.

Program / Code

```
import pandas as pd

data = {
"school": ["S1", "S2", "S3", "S4", "S5", "S6"],
"school_code": ["s001", "s002", "s003", "s001", "s002", "s004"],
"class": ["V", "V", "VI", "VI", "V", "VI"],
"name": [
"Alberto Franco", "Gino Mcneill", "Ryan Parkes",
"Eesha Hinton", "Gino Mcneill", "David Parkes"
],
"age": [12, 13, 13, 13, 14, 12]
}

df = pd.DataFrame(data)

result = df.groupby("school_code")["age"].agg(
["mean", "min", "max"]
)

print(result)
```

Output

```
             mean  min  max
school_code
s001         12.5   12   13
s002         13.5   13   14
s003         13.0   13   13
s004         12.0   12   12
```

EXPERIMENT 18

Question: Write a Pandas program to split the following given dataframe into groups based on school code and class.

Program / Code

```
import pandas as pd

data = {
"school": ["S1", "S2", "S3", "S4", "S5", "S6"],
"school_code": ["s001", "s002", "s003", "s001", "s002", "s004"],
"class": ["V", "V", "VI", "VI", "V", "VI"],
"name": [
"Alberto Franco", "Gino Mcneill", "Ryan Parkes",
"Eesha Hinton", "Gino Mcneill", "David Parkes"
],
"age": [12, 13, 13, 13, 14, 12]
}

df = pd.DataFrame(data)

grouped = df.groupby(["school_code", "class"])

for key, group in grouped:
print("\nGroup:", key)
print(group)
```

Output

```
Group: ('s001', 'V')
  school school_code class            name  age
0     S1        s001     V  Alberto Franco   12

Group: ('s001', 'VI')
  school school_code class          name  age
3     S4        s001    VI  Eesha Hinton   13

Group: ('s002', 'V')
  school school_code class          name  age
1     S2        s002     V  Gino Mcneill   13
4     S5        s002     V  Gino Mcneill   14

Group: ('s003', 'VI')
  school school_code class         name  age
2     S3        s003    VI  Ryan Parkes   13

Group: ('s004', 'VI')
  school school_code class          name  age
5     S6        s004    VI  David Parkes   12
```

EXPERIMENT 19

Question: Write a Pandas program to display the dimensions or shape of the World alcohol consumption dataset. Also extract the column names from the dataset.

Program / Code

```
import pandas as pd

data = {
"Year": [1986, 1986, 1985, 1986, 1987],
"WHO_region": [
"Western Pacific", "Americas", "Africa",
"Americas", "Americas"
],
"Country": [
"Viet Nam", "Uruguay", "Cte d'Ivoire",
"Colombia", "Saint Kitts and Nevis"
],
"Beverage_Types": [
"Wine", "Other", "Wine", "Beer", "Beer"
],
"Display_Value": [0.00, 0.50, 1.62, 4.27, 1.98]
}

df = pd.DataFrame(data)

print("Shape:")
print(df.shape)

print("\nColumn names:")
print(df.columns.tolist())
```

Output

```
Shape:
(5, 5)

Column names:
['Year', 'WHO_region', 'Country', 'Beverage_Types', 'Display_Value']
```

EXPERIMENT 20

Question: Write a Pandas program to find the index of a given substring of a DataFrame column.

Program / Code

```
import pandas as pd

df = pd.DataFrame({
"name": [
"Alberto Franco",
"Gino Mcneill",
"Ryan Parkes",
"Eesha Hinton",
"David Parkes"
]
})

df["index"] = df["name"].str.find("Parkes")

print(df)
```

Output

	name	index
0	Alberto Franco	-1
1	Gino Mcneill	-1
2	Ryan Parkes	5
3	Eesha Hinton	-1
4	David Parkes	6